

RODENTS

Ranging from tiny mice to animals the size of a pig, rodents live in almost every habitat type. They make up nearly half of all mammal species.

Members of the order Rodentia are distinguished by having a pair of prominent upper and lower incisor teeth (often orange or yellow in colour), which continue to grow throughout life. Gnawing – a characteristic behaviour of all rodents – wears the teeth away at the same rate as they grow. Rodents have no canine teeth, just a long space between the incisors and the three or four cheek teeth in each jaw. The scientific classification of rodents is also based on other features of their teeth and jaws that are not visible externally.

SPECIES DIVERSITY

To suit their various lifestyles, rodents often have special adaptations such as webbed toes, big ears, or long hind feet for a bounding gait. Some rodents burrow, others live in trees or in the water, but none live in the sea. Many species inhabit desert regions, where they may never drink, deriving all the water they need from food.

THE IMPACT OF RODENTS

Several rodent species carry fatal diseases that have killed millions of people. Others consume or contaminate huge quantities of stored food meant for humans. The house mouse has become the world's most widespread wild mammal through its close association with humans. This species inhabits every continent, except Antarctica, even surviving in mines and cold stores. Some rodents cause damage to crops or trees, or burrow in inconvenient places. Beavers can transform entire habitats, affecting hundreds of other animal and plant species.

On the positive side, rodents are a vital food source for predatory animals and, in some countries, for humans as well. Many small rodents, such as hamsters, are specially bred as pets.

PHYLUM	CHORDATA
CLASS	MAMMALIA
ORDER	RODENTIA
FAMILIES	34
SPECIES	About 2,500

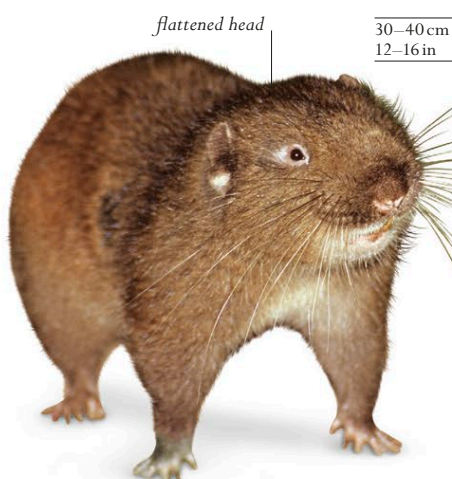
DEBATE

RODENT ORGANIZATION

The huge diversity of species in the order Rodentia makes the task of classifying them rather complicated. With 34 families of modern rodent to organize, experts in zoological classification have divided them into two subgroups, according to differences in their skulls, teeth, and jaws. The Sciurognathi (squirrel-jaws) are globally widespread, and include all squirrels, beavers, and mouse-like rodents. The Hystricognathi (porcupine-jaws) includes the cavies, porcupines, chinchillas, and the capybara, and these are mostly restricted to the southern hemisphere and the tropics.

MOUNTAIN BEAVER

Only one species of the once widespread family Aplodontiidae has survived into modern times. The mountain beaver, or sewellel, lives in burrows in the damp forests and plantations of western North America.



MOUNTAIN BEAVER

Aplodontia rufa
This most primitive type of living rodent inhabits forested coastal mountains of W. Canada and the USA.

SQUIRRELS AND CHIPMUNKS

Occurring almost everywhere, except in the polar regions, Australia, and the Sahara Desert, members of the family Sciuridae range from tropical rainforests to Arctic tundra, and from the tops of trees to underground tunnels. This family includes typical bushy-tailed tree squirrels, many species of burrow-digging ground squirrels, chipmunks, and marmots. They feed mainly on nuts and seeds.

AMERICAN RED SQUIRREL

Tamiasciurus hudsonicus
The chattering calls and sharp barks of this red squirrel are common sounds of coniferous woodlands in Canada and the N. USA.



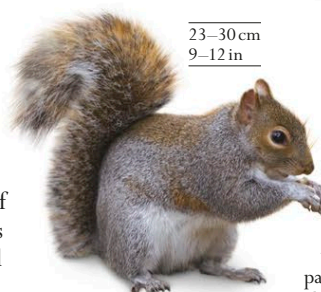
17–20 cm
6½–8 in

PALE GIANT SQUIRREL

Ratufa affinis
Of the four species of giant squirrel, this species inhabits the forests of the Malay Peninsula, Borneo, and Sumatra.



30–38 cm
12–15 in



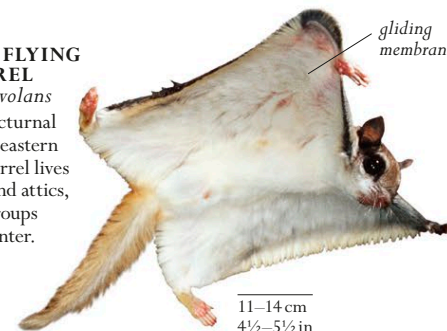
23–30 cm
9–12 in

GREY SQUIRREL

Sciurus carolinensis
Common in the E. USA, this species was introduced to parts of Europe, where it is slowly displacing the native red squirrel.

SOUTHERN FLYING SQUIRREL

Glaucomys volans
A strictly nocturnal native of the eastern USA, this squirrel lives in tree holes and attics, often in groups during winter.



11–14 cm
4½–5½ in



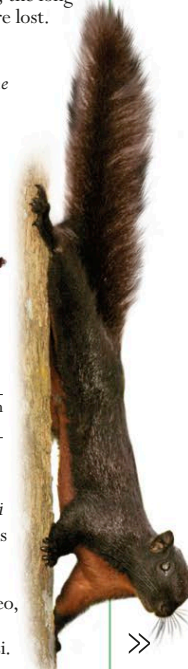
32–39 cm
12½–15½ in

GRIZZLED GIANT SQUIRREL

Ratufa macroura
One of the world's largest tree squirrels, this species occurs in S. India and Sri Lanka. It eats fruits, flowers, and insects.

EURASIAN RED SQUIRREL

Sciurus vulgaris
When this squirrel moults into its summer coat, the long hairs on its ears are lost.



PREVOST'S SQUIRREL

Callosciurus prevostii
There are 17 subspecies of these tricoloured squirrels. Prevost's is found in Malaysia, Borneo, Sumatra, and nearby islands, such as Sulawesi.